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Inventor(s)

LI; Jiaqi et al.

HEAT EXCHANGE ASSEMBLY, BATTERY AND ELECTRICAL DEVICE

Abstract

A heat exchange assembly, a battery and an electrical device are disclosed. The heat exchange assembly includes a current collecting piece and a plurality of heat exchange pieces. The heat exchange piece is constructed with a first flow channel and a first joint portion communicating with the first flow channel. The current collecting piece is constructed with a second flow channel and a plurality of second joint portions. The plurality of second joint portions each communicate with the second flow channel and are configured to be connected to the first joint portions of the plurality of heat exchange pieces respectively, such that the second flow channel of the current collecting piece communicates with the first flow channels of the plurality of heat exchange pieces. When the heat exchange assembly and battery units form a group, the assembly procedure is simple, and the assembly and fabrication efficiency is high.

Inventors: LI; Jiaqi (Ningde, CN), HUANG; Yincheng (Ningde, CN), You; Kaijie (Ningde, CN), LIU; Chen (Ningde, CN), SHEN; Zhen (Ningde, CN)

Applicant: CONTEMPORARY AMPEREX TECHNOLOGY (HONG KONG) LIMITED
(Hong Kong, CN)

Family ID: 87880342

Assignee: CONTEMPORARY AMPEREX TECHNOLOGY (HONG KONG) LIMITED
(Hong Kong, CN)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International application PCT/CN2023/128221 filed on Oct. 31, 2023 that claims priority to Chinese Patent Application No. 202223090680.X filed on Nov. 21, 2022, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This application belongs to the technical field of battery heat exchange, and particularly relates to a heat exchange assembly, a battery and an electrical device.

BACKGROUND

[0003] A heat exchange assembly, as a core component of a battery thermal management system, usually exchanges heat with a battery unit, so as to play a role in heating and cooling the battery unit to ensure that the battery unit works within an appropriate temperature range and to ensure stable running of a battery.

[0004] However, in an actual manufacturing process, the fabrication operation between the battery unit and the heat exchange assembly is complicated, and the assembly efficiency is relatively low.

[0005] The description in the background section merely provides background information related to this application, but does not necessarily constitute prior art or related art.

SUMMARY

[0006] An objective of embodiments of this application is to provide a heat exchange assembly, a battery and an electrical device to solve problems including but not limited to complicated fabrication operation of a battery unit and a heat exchange assembly and relatively low assembly efficiency.

[0007] A technical solution disclosed in an embodiment of this application is as follows:

[0008] According to a first aspect, a heat exchange assembly is provided. The heat exchange assembly includes a current collecting piece and a plurality of heat exchange pieces. The heat exchange piece is constructed with a first flow channel and a first joint portion communicating with the first flow channel. The current collecting piece is constructed with a second flow channel and a plurality of second joint portions. The plurality of second joint portions each communicate with the second flow channel and are configured to be connected to the first joint portions of the plurality of heat exchange pieces respectively, such that the second flow channel of the current collecting piece communicates with the first flow channels of the plurality of heat exchange pieces.

[0009] According to the heat exchange assembly in this embodiment of this application, when the heat exchange assembly is assembled with battery units, the plurality of battery units are combined

with the plurality of heat exchange pieces first, and then the plurality of second joint portions on the current collecting piece are connected to the first joint portions on the heat exchange pieces in one-to-one correspondence, thereby completing assembly of the battery units and the heat exchange assembly. The heat exchange assembly may simultaneously connect the plurality of heat exchange pieces to the current collecting piece through the second joint portions by arranging the plurality of second joint portions on one current collecting piece, which, compared with a form of separate connection of a plurality of current collecting pieces, can save parts and reduce assembly procedures, thereby improving the assembly and fabrication efficiency.

[0010] In an embodiment, the current collecting piece includes a connecting face oriented toward the heat exchange pieces, and the plurality of second joint portions are disposed on the connecting face.

[0011] When the heat exchange assembly according to this embodiment of this application is assembled, the connecting face of the current collecting piece is placed toward the heat exchange pieces, and the second joint portions are disposed opposite to the heat exchange pieces, thereby facilitating the connection of the first joint portions and the second joint portions, and also contributing to the improvement of the assembly efficiency.

[0012] In an embodiment, the plurality of heat exchange pieces are spaced apart along a first direction, and the plurality of second joint portions are spaced out on the connecting face along the first direction, such that the plurality of second joint portions are in one-to-one correspondence with the first joint portions of the plurality of heat exchange pieces.

[0013] When the heat exchange assembly according to this embodiment of this application is assembled with the battery units, the plurality of second joint portions on the current collecting piece are adjusted to be distributed along the first direction and opposite to the first joint portions, such that the first joint portions may be aligned with the second joint portions, which makes the connecting operation of a heat exchange piece and the current collecting piece simpler and quicker, and also contributes to the improvement of the assembly efficiency.

[0014] In an embodiment, the heat exchange assembly further includes connectors, and each connector is disposed at a junction of the first joint portion and the second joint portion to connect the first joint portion to the second joint portion.

[0015] According to the heat exchange assembly in this embodiment of this application, the first joint portion and the second joint portion are connected through the connector, such that the connecting operation of the first joint portion and the second joint portion is simple and quick, which contributes to the improvement of the assembly efficiency of the heat exchange assembly and the battery units.

[0016] In an embodiment, the first joint portion includes a first through hole, the second joint portion includes a second through hole, the connector includes a first end and a second end that extend along a connecting direction of the first joint portion and the second joint portion, the first end is connected into the first through hole, the second end is connected into the second through hole, and the connector includes a connecting hole running through the first end and the second end.

[0017] According to the heat exchange assembly in this embodiment of this application, after the first end and the second end of the connector are connected into the first through hole and the second through hole respectively, the connection, fixing and communication of the current collecting piece and the heat exchange piece may be simultaneously realized, thereby simplifying assembly operation, which contributes to the improvement of the assembly efficiency of the heat exchange assembly and the battery units.

[0018] In an embodiment, the first end is inserted in the first through hole, and the second end is inserted in the second through hole.

[0019] According to the heat exchange assembly in this embodiment of this application, the first end is inserted in the first through hole, and the second end is inserted in the second through hole,

such that the connector is connected to the current collecting piece and the heat exchange piece respectively. The connecting operation is simple, which contributes to the improvement of the assembly efficiency of the heat exchange assembly and the battery units.

[0020] In an embodiment, the heat exchange assembly further includes first seals, and the first seal is disposed between the connector and a hole wall of the first through hole; and/or, the first seal is disposed between the connector and a hole wall of the second through hole.

[0021] According to the heat exchange assembly in this embodiment of this application, the first seals reduce the risk of leakage of a heat exchange medium, which contributes to the improvement of the heat exchange efficiency.

[0022] In an embodiment, heat exchange piece includes a first side plate and a second side plate, and the first side plate and the second side plate are connected and close in to form the first flow channel and the first through hole.

[0023] The heat exchange piece according to this embodiment of this application is formed by the connecting and closing in of the first side plate and the second side plate, and is simple in structure and also convenient to machine and manufacture. In addition, in a specific manufacturing process, the first side plate and the second side plate may be manufactured by stamping, such that the machining and manufacturing of the heat exchange piece are simpler, and the manufacturing cost is lower.

[0024] In an embodiment, the connector includes a first connecting portion and a second connecting portion, the first connecting portion is connected to the first side plate, the second connecting portion is connected to the second side plate, and the first connecting portion and the second connecting portion close in to form the connecting hole.

[0025] According to the heat exchange assembly in this embodiment of this application, the connector uses a separate structure of the first connecting portion and the second connecting portion, and the first connecting portion and the second connecting portion may be manufactured by stamping, or the like, such that the machining and manufacturing of the connector are simpler.

[0026] In an embodiment, the first side plate and the first connecting portion are of a one-piece structure; or the first connecting portion is bonded to the first side plate; or the first connecting portion is welded to the first side plate.

[0027] The heat exchange pieces in the heat exchange assembly according to this embodiment of this application are easier to manufacture and machine, and the machining and manufacturing cost is low.

[0028] In an embodiment, the second side plate and the second connecting portion are of a one-piece structure; or the second connecting portion is bonded to the second side plate; or the second connecting portion is welded to the second side plate.

[0029] The heat exchange pieces in the heat exchange assembly according to this embodiment of this application are easier to manufacture and machine, and the machining and manufacturing cost is low.

[0030] In an embodiment, the second end is threaded in the second through hole, and the first end is screwed in the first through hole, such that the heat exchange piece is connected to the current collecting piece.

[0031] According to the heat exchange assembly in this embodiment of this application, the second end of the connector is threaded in the second through hole, and the first end of the connector is screwed in the first through hole. By threaded connection, on the one hand, the connecting operation is simple, which contributes to the improvement of the assembly effect; and in addition, the manner of threaded connection has good connection reliability and also ensures stable connection between the connector and the heat exchange piece. Moreover, threaded connection has a sealing effect, which can also reduce the leakage of a heat exchange medium and improve the heat exchange effect.

[0032] In an embodiment, the current collecting piece includes a first plate body and a second plate

body, and the first plate body and the second plate body are connected and close in to form the second flow channel; and the second through holes are disposed on the first plate body.

[0033] According to the heat exchange assembly in this embodiment of this application, when the first joint portion is connected to the second joint portion, the first end of the connector passes through the second through hole first, and then is threadedly locked in the first through hole to fix the first plate body on the heat exchange piece, and finally the second plate body is fixed on the first plate body, thereby realizing assembly of the heat exchange pieces and the current collecting piece. The assembly operation is simple, convenient and quick, which contributes to the improvement of the heat exchange assembly and the battery units.

[0034] In an embodiment, a second seal is sandwiched between the first plate body and the second plate body, and the second seal surrounds the second flow channel; and/or, one of the first plate body and the second plate body is provided with an annular seal bulge, and the other of the first plate body and the second plate body is provided with an annular seal groove; and the annular seal bulge is inserted in the annular seal groove and surrounds the second flow channel.

[0035] The heat exchange assembly according to this embodiment of this application can seal the second flow channel, reduce the risk of leakage of the heat exchange medium, and improve the heat exchange efficiency.

[0036] In an embodiment, the second joint portion further includes a first bulge, the first bulge protrudes from the connecting face, and the second through hole runs through the first bulge along the connecting direction of the first joint portion and the second joint portion.

[0037] According to the heat exchange assembly in this embodiment of this application, the disposing of the first bulge can increase the thickness of the second through hole, improve the structural strength of the second through hole, and improve the connection reliability of the current collecting piece and the connector. In addition, the disposing of first bulge can also increase the depth of the second through hole, thereby increasing the connection area of the connector and the second through hole, improving the connection reliability of the current collecting piece and the connector, and in turn, improving the structural strength of the entire heat exchange assembly.

[0038] In an embodiment, the first joint portion further includes a second bulge, the second bulge is disposed on an outer surface of the heat exchange piece oriented toward the current collecting piece, the first through hole runs through the second bulge along the connecting direction of the first joint portion and the second joint portion, and the first bulge abuts against the second bulge.

[0039] According to the heat exchange assembly in this embodiment of this application, the disposing of the second bulge can increase the thickness of the first through hole, improve the structural strength of the first through hole, improve the connection reliability of the heat exchange piece and the connector, and in turn, improve the structural strength of the entire heat exchange assembly.

[0040] In an embodiment, a third seal is sandwiched between the first bulge and the second bulge, and the connector is threaded in the third seal.

[0041] The heat exchange assembly according to this embodiment of this application reduces the risk of leakage of the heat exchange medium in the second flow channel, reduces the leakage of the heat exchange medium in the heat exchange assembly, and contributes to the improvement of the heat exchange efficiency.

[0042] In an embodiment, the second joint portion further includes a third bulge, the third bulge is disposed on an inner wall of the second flow channel close to the heat exchange piece, and the second through hole runs through the third bulge along the connecting direction of the first joint portion and the second joint portion.

[0043] According to the heat exchange assembly in this embodiment of this application, the disposing of the third bulge can increase the thickness of the second through hole, improve the structural strength of the second through hole, and improve the connection reliability of the current collecting piece and the connector. In addition, the disposing of the third bulge can also increase the

depth of the second through hole, thereby increasing the connection area of the connector and the second through hole, improving the connection reliability of the current collecting piece and the connector, and in turn, improving the structural strength of the entire heat exchange assembly.

[0044] According to a second aspect, a battery is provided. The battery includes the foregoing heat exchange assembly.

[0045] The battery according to an embodiment of this application uses the foregoing heat exchange assembly. The assembly operation of the heat exchange assembly and the battery units is simple, and the assembly efficiency is high, which contributes to the improvement of the production efficiency of the battery and the reduction of the manufacturing cost of the battery.

[0046] According to a third aspect, an electrical device is provided. The electrical device includes the foregoing battery.

[0047] The electrical device according to an embodiment of this application uses the foregoing battery. The manufacturing cost of the battery is low, which contributes to the reduction of the manufacturing cost of the electrical device.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0048] To describe the technical solutions in some embodiments of this application more clearly, the following outlines the drawings to be used in the description of some exemplary embodiments of this application. Evidently, the drawings outlined below are merely some embodiments of this application. A person of ordinary skill in the art may derive other drawings from such drawings without making any creative efforts.

[0049] FIG. 1 is a schematic structural diagram of a vehicle provided by an embodiment of this application.

[0050] FIG. 2 is a schematic structural exploded view of a battery according to an embodiment of this application.

[0051] FIG. 3 is a schematic structural exploded view of a battery cell according to an embodiment of this application.

[0052] FIG. 4 is a schematic structural diagram of a heat exchange assembly and battery units provided by an embodiment of this application after being assembled.

[0053] FIG. 5 is a schematic structural exploded view of the heat exchange assembly shown in FIG. 1.

[0054] FIG. 6 is a schematic structural diagram of the heat exchange assembly provided by an embodiment of this application as observed from a viewing angle.

[0055] FIG. 7 is a schematic structural exploded view of the heat exchange assembly shown in FIG. 6.

[0056] FIG. 8 is a schematic structural diagram of the heat exchange assembly shown in FIG. 6 as observed from another viewing angle.

[0057] FIG. 9 is a sectional view along an A-A line in FIG. 8.

[0058] FIG. 10 is a close-up view of a part B in FIG. 9.

[0059] FIG. 11 is a schematic structural partially exploded view of the heat exchange assembly provided by an embodiment of this application.

[0060] FIG. 12 is a close-up view of a part C in FIG. 11.

[0061] FIG. 13 is a schematic structural partially exploded view of the heat exchange assembly provided by an embodiment of this application.

[0062] FIG. 14 is a schematic structural diagram of a current collecting piece provided by an embodiment of this application.

[0063] FIG. 15 is a schematic structural diagram of the heat exchange assembly provided by an

embodiment of this application.

[0064] FIG. **16** is a sectional view along a D-D line in FIG. **15**.

[0065] FIG. **17** is a close-up view of a part E in FIG. **16**.

[0066] FIG. **18** is a schematic structural exploded view of the heat exchange assembly shown in FIG. **15** as observed from a viewing angle.

[0067] FIG. **19** is a schematic structural exploded view of the heat exchange assembly shown in FIG. **15** as observed from another viewing angle.

[0068] FIG. **20** is a schematic structural partially exploded view of the heat exchange assembly shown in FIG. **15** as observed from a viewing angle.

[0069] FIG. **21** is a schematic structural partially exploded view of the heat exchange assembly shown in FIG. **15** as observed from another viewing angle.

LIST OF REFERENCE NUMERALS

[0070] **1000.** vehicle; **1100.** battery; **1200.** controller; **1300.** motor; **10.** box; **11.** first portion; **12.** second portion; **20.** battery cell; **21.** end cap; **22.** shell; **23.** electrode assembly;

[0071] **1.** heat exchange assembly; **100.** current collecting piece; **110.** first plate body; **111.** annular seal bulge; **112.** annular abutting bulge; **120.** second plate body; **121.** annular seal groove; **122.** annular accommodation groove; **101.** second flow channel; **102.** second joint portion; **1021.** second through hole; **1022.** first bulge; **1023.** third bulge; **103.** connecting face; **200.** heat exchange piece; **210.** first side plate; **211.** connecting bulge; **212.** first semi-ring bulge; **220.** second side plate; **221.** second semi-ring bulge; **201.** first flow channel; **202.** first joint portion; **2021.** first through hole; **2022.** second bulge; **203.** accommodation clearance; **204.** seal accommodation groove; **300.** connector; **310.** first connecting portion; **320.** second connecting portion; **301.** connecting hole; **410.** first seal; **420.** second seal; **430.** third seal; **440.** second screw; **500.** battery unit; **610.** liquid inlet joint; and **620.** liquid outlet joint.

DETAILED DESCRIPTION OF EMBODIMENTS

[0072] The following describes embodiments of this application in detail. Examples of the embodiments are shown in FIGS. **1** to **21**, throughout which the same or similar reference numerals represent the same or similar components or components of the same or similar functions. The embodiments described below with reference to FIGS. **1** to **21** are exemplary, and are intended to construe this application but not to limit this application.

[0073] Understandably, in the description of this application, a direction or positional relationship indicated by the terms such as “length”, “width”, “up”, “down”, “front”, “back”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “in”, and “out” is a direction or positional relationship based on the illustration in the drawings, and is merely intended for ease or brevity of description of this application, but does not necessarily mean or imply that the indicated device or component is located in the specified direction or constructed or operated in the specified direction. Therefore, such terms are not to be understood as a limitation on this application.

[0074] In addition, the terms “first” and “second” are used merely for descriptive purposes but are not to be construed as indicating or implying relative importance or implicitly specifying the quantity of technical features indicated. Therefore, a feature defined with “first” or “second” may explicitly or implicitly include one or more of the features. In the description of this application, unless otherwise expressly and specifically defined, “a plurality of” means two or more.

[0075] In this application, unless otherwise expressly specified or defined, the terms such as “mount”, “concatenate”, “connect” and “fix” shall be understood broadly, for example, mean a fixed connection, a detachable connection, or a one-piece configuration; or mean a mechanical connection or an electrical connection; or mean a direct connection or an indirect connection implemented through an intermediary; or mean internal communication between two components or interaction between two components. A person of ordinary skill in the art may understand specific meaning of the above terms in this application based on specific circumstances.

[0076] It is hereby noted that, in the description of this application, the term “and/or” merely

indicates a relationship between related items, and represents three possible relationships. For example, “A and/or B” may represent the following three circumstances: A alone, both A and B, and B alone.

[0077] Further, it is hereby noted that in embodiments of this application, the same component or the same part is denoted by the same reference numeral in the accompanying drawings. For identical components or parts in embodiments of this application, the drawings may illustrate the reference numeral of only one of the components or parts as an example. Understandably, the reference numeral is equally applicable to other identical components or parts.

[0078] In this application, the terms such as “an embodiment”, “some embodiments”, “examples”, “specific examples” or “some examples” mean that the specific feature, structure, material or characteristic described in combination with the embodiment or example is included in at least one embodiment or example of this application. In the specification, the exemplary expressions of the above terms do not necessarily refer to the same embodiment or example. Moreover, the described specific feature, structure, material or characteristic may be combined in an appropriate manner in any one or more embodiments or examples. In addition, a person skilled in the art may integrate or combine different embodiments or examples as well as the features of different embodiments or examples described herein to the extent that they do not contradict each other.

[0079] Currently, the market development trend shows that, power batteries are applied more extensively. The power batteries are not only applied to energy storage power systems such as hydro, thermal, wind, and solar power stations, but also widely applied to electric means of transport such as electric bicycles, electric motorcycles, and electric vehicles, as well as in various fields such as military equipment and aerospace.

[0080] A battery cannot work normally when the temperature is too high or too low, and may even cause a safety accident. Therefore, it is necessary to perform temperature control on the battery. In some circumstances, the battery is internally provided with a heat exchange assembly through which a battery unit is heated and cooled, thereby realizing temperature control over the battery. The heat exchange assembly usually includes a plurality of heat exchange pieces and a plurality of current collecting pieces. In the process of assembling the battery unit with the heat exchange assembly, one heat exchange piece and one battery unit are stacked and assembled first, and then two current collecting pieces are mounted at two ends of a heat exchange piece respectively to obtain a heat exchange module. Then, a plurality of heat exchange modules are assembled together one by one. Moreover, in the assembly process, when two adjacent heat exchange modules are assembled, it is also necessary to connect current collecting pieces on the two adjacent heat exchange modules together with the aid of auxiliary parts, thereby realizing communication of the heat exchange modules. The applicant hereof has noticed that, the plurality of current collecting pieces in the heat exchange assembly are in a structural form of separate connection along an arrangement direction of the heat exchange pieces, which requires the heat exchange modules to be assembled one by one and with the aid of the auxiliary parts, such that the assembly operation is cumbersome, and the assembly efficiency is low.

[0081] In order to alleviate the problem, the applicant has designed a heat exchange assembly after in-depth research. When the heat exchange assembly is assembled with battery units, the plurality of battery units are combined with a plurality of heat exchange pieces first, and then a plurality of second joint portions on a current collecting piece are connected to first joint portions on the heat exchange pieces in one-to-one correspondence, thereby completing assembly of the battery units and the heat exchange assembly. The heat exchange assembly may simultaneously connect the plurality of heat exchange pieces to the current collecting piece through the second joint portions by arranging the plurality of second joint portions on one current collecting piece, which, compared with the form of separate connection of the plurality of current collecting pieces, can save parts and reduce assembly procedures, thereby improving the assembly and fabrication efficiency.

[0082] An embodiment of this application further provides an electrical device using a battery as a

power supply. The electrical device may be, but is not limited to, a mobile phone, a tablet, a laptop computer, an electric toy, an electric tool, an electric power cart, an electric vehicle, a ship, a spacecraft, or the like. The electric toy may include stationary or mobile electric toys, such as a game console, an electric car toy, an electric ship toy, and an electric airplane toy. The spacecraft may include an airplane, a rocket, a space shuttle, a spaceship, and the like.

[0083] For ease of description in the following embodiments, a vehicle **1000** is used as an example of an electrical device according to an embodiment of this application.

[0084] Referring to FIG. 1, FIG. 1 is a schematic structural diagram of a vehicle **1000** provided by some embodiments of this application. The vehicle **1000** may be an oil-fueled vehicle, a natural gas vehicle, or a new energy vehicle. The new energy vehicle may be a battery electric vehicle, a hybrid electric vehicle, a range-extended electric vehicle, or the like. A battery **1100** is disposed inside the vehicle **1000**. The battery **1100** may be disposed at the bottom, front, or rear of the vehicle **1000**. The battery **1100** may be configured to supply power to the vehicle **1000**. For example, the battery **1100** may serve as an operating power supply of the vehicle **1000**. The vehicle **1000** may further include a controller **1200** and a motor **1300**. The controller **1200** is configured to control the battery **1100** to supply power to the motor **1300**, for example, to meet the working power requirements of the vehicle **1000** in start, navigation and running.

[0085] In some embodiments of this application, the battery **1100** serves not only as an operating power supply of the vehicle **1000**, but may also serve as a driving power supply of the vehicle **1000** to provide driving power for the vehicle **1000** in place of or partly in place of fuel oil or natural gas.

[0086] Referring to FIG. 2, in an embodiment of the battery **1100**, the battery **1100** includes a box **10** and a battery unit **500**. The battery unit **500** is accommodated in the battery box **10**. The battery box **10** is configured to provide an accommodation space for the battery unit **500**, and the battery box **10** may assume various structures. In some embodiments, the battery box **10** may include a first portion **11** and a second portion **12**. The first portion **11** and the second portion **12** fit and cover each other. The first portion **11** and the second portion **12** jointly define the accommodation space for accommodating the battery unit **500**. The second portion **12** may be of a hollow structure opened at one end. The first portion **11** may be of a plate-like structure. The first portion **11** fits and covers an opening side of the second portion **12**, such that the first portion **11** and the second portion **12** jointly define the accommodation space. The first portion **11** and the second portion **12** may be each of a hollow structure opened at one end. An opening side of the first portion **11** fits and covers the opening side of the second portion **12**. Certainly, the battery box **10** formed by the first portion **11** and the second portion **12** may have various shapes, such as a cylinder and a cuboid.

[0087] In the battery **1100**, there may be a plurality of battery units **500**. The plurality of battery units **500** may be connected in series or in parallel or in parallel and series. Connected in parallel and series means that there are both series and parallel connections in the plurality of battery units **500**. The battery unit **500** may be a battery cell **20** or a battery module, where the battery module refers to a modular component that is formed by assembling a plurality of battery cells **20**.

[0088] In an embodiment, the plurality of battery cells **20** may be directly connected together in series or in parallel or in parallel and series, and then the entirety of the plurality of battery cells **20** is accommodated in the box **10**. Certainly, the plurality of battery cells **20** may be connected in series or in parallel or in parallel and series to form a battery module first, and then a plurality of battery modules are connected in series or in parallel or in parallel and series to form an entire battery **1100** and accommodated in the box **10**. The battery **1100** may further include other structures. For example, the battery **1100** may further include a busbar component for realizing an electrical connection between the plurality of battery cells **20**.

[0089] Each battery cell **20** may be a secondary battery or a primary battery, and may also be a lithium-sulfur battery, a sodium-ion battery, or a magnesium-ion battery, but is not limited thereto. The battery cells **20** may be cylindrical, flat, rectangular, or other shapes, etc.

[0090] In another embodiment of the battery **1100**, the battery **1100** does not include the box **10**, but a plurality of battery cells **20** are electrically connected to form a whole by means of a necessary fixing structure, and then fitted into an electrical device.

[0091] Referring to FIG. **3**, FIG. **3** is a schematic structural exploded view of a battery cell **20** provided by some embodiments of this application. The battery cell **20** is the smallest unit that makes up a battery. As shown in FIG. **3**, the battery cell **20** includes an end cap **21**, a shell **22**, an electrode assembly **23**, and other functional components.

[0092] The end cap **21** is a component that fits and covers an opening of the shell **22** to isolate an internal environment of the battery cell **20** from an external environment. Without limitation, the shape of the end cap **21** may be adapted to the shape of the shell **22** to fit the shell **22**. In an embodiment, the end cap **21** may be made of a material having a certain degree of hardness and strength (e.g., aluminum alloy). In this way, the end cap **21** is less likely to deform when subjected to extrusion and collision, allowing the battery cell **20** to have higher structural strength and improved safety performance. The end cap **21** may be provided with functional components such as electrode terminals. The electrode terminals may be configured to be electrically connected to the electrode assembly **23** to output or input electrical energy of the battery cell **20**. In some embodiments, the end cap **21** may further be provided with a pressure relief mechanism configured to release an internal pressure when the internal pressure or temperature of the battery cell **20** reaches a threshold value. The end cap **21** may also be made of a variety of materials such as copper, iron, aluminum, stainless steel, aluminum alloy, and plastic, which are not particularly limited in these embodiments of this application. In some embodiments, an insulator may further be disposed on an inner side of the end cap **21**, and the insulator may be configured to isolate an electrically connected component in the shell **22** from the end cap **21** to reduce the risk of a short circuit. Exemplarily, the insulator may be plastic, rubber, or the like.

[0093] The shell **22** is an assembly configured to fit the end cap **21** to form the internal environment of the battery cell **20**. The formed internal environment may be configured to accommodate the electrode assembly **23**, an electrolyte solution, and other components. The shell **22** and the end cap **21** may be stand-alone components. The opening may be disposed on the shell **22**. At the opening, the end cap **21** fits and covers the opening to form the internal environment of the battery cell **20**. Without limitation, the top cap **21** and the shell **22** may be integrated. The end cap **21** and the shell **22** may form a common connecting face before other components are put into the shell. Subsequently, when an interior of the shell **22** needs to be sealed, the end cap **21** is made to fit and cover the shell **22**. The shell **22** may have various shapes and sizes, such as a cuboid, a cylinder, or a hexagonal prism. The shape of the shell **22** may be determined depending on the specific shape and size of the electrode assembly **23**. The shell **22** may be made of a variety of materials, such as copper, iron, aluminum, stainless steel, aluminum alloy, or plastic, which are not particularly limited in these embodiments of this application.

[0094] The electrode assembly **23** is a component that reacts electrochemically in the battery cell **20**. The shell **22** may contain one or more electrode assemblies **23**. The electrode assembly **23** is primarily formed by winding or stacking a positive electrode plate and a negative electrode plate, and a separator is usually disposed between the positive electrode plate and the negative electrode plate. Parts of the positive electrode plate and the negative electrode plate having active materials constitute a body portion of the electrode assembly **23**, while parts of the positive electrode plate and the negative electrode plate having no active material constitute tabs respectively. The positive tab and the negative tab may be located at one end of the body portion together or at two ends of the body portion respectively. In a charge-and-discharge process of the battery, the positive active material and the negative active material react with the electrolyte solution, and the tabs are connected to the electrode terminals to form a current circuit.

[0095] With reference to FIG. **4** and FIG. **5**, a plurality of heat exchange pieces **200** and a plurality of groups of battery units **500** are alternately spaced out. Each group of battery units **500** is located

between two adjacent heat exchange pieces **200**. The two adjacent heat exchange pieces **200** exchange heat with two opposite sides of each group of battery units **500**, such that double-sided heat exchange of the battery units **500** is realized. By increasing the heat exchange area, the battery is quickly heated and cooled. The two adjacent heat exchange pieces **200** are directly in contact with two opposite side faces of the battery units **500**, or bonded thereto through a heat-conducting adhesive, such that the battery units **500** exchange heat with the heat exchange pieces **200**, and double-sided water cooling of the battery is realized, thereby solving the problem of an excessively high temperature rise in the battery and the problem of an excessively low temperature in the battery. After the battery units **500** and heat exchange assembly **1** in a group are put into the box, a part of the heat exchange pieces **200** in the heat exchange assembly **1** may further serve as cross beams of the box **10** to improve the structural strength of the battery. Each group of battery units **500** may include one or more battery units **500**, and the specific number thereof may be selected according to actual needs, and is not limited herein.

[0096] With reference to FIG. 7, in an embodiment of this application, a heat exchange assembly **1** is provided. The heat exchange assembly **1** includes a current collecting piece **100** and a plurality of heat exchange pieces **200**. With reference to FIG. 8, FIG. 9 and FIG. 10 together, the heat exchange piece **200** is constructed with a first flow channel **201** and a first joint portion **202** communicating with the first flow channel **201**. The current collecting piece **100** is constructed with a second flow channel **101** and a plurality of second joint portions **102**. The plurality of second joint portions **102** each communicate with the second flow channel **101** and are configured to be connected to the first joint portions **202** of the plurality of heat exchange pieces **200** respectively, such that the second flow channel **101** of the current collecting piece **100** communicates with the first flow channels **201** of the plurality of heat exchange pieces **200**.

[0097] The heat exchange piece **200** refers to a component that exchanges heat with the battery unit **500**. The heat exchange piece **200** has good heat-conducting performance to exchange heat with the battery unit **500**. The heat exchange piece **200** may be made of materials having good heat-conducting performance, such as aluminum and copper.

[0098] The first flow channel **201** refers to a channel that is disposed in the heat exchange piece **200** and available for flow of a heat exchange medium.

[0099] The first joint portion **202** refers to a portion on the heat exchange piece **200** for being connected to the current collecting piece **100**.

[0100] The current collecting piece **100** refers to a component that introduces the collected heat exchange medium into the heat exchange pieces **200** or that is configured to collect the heat exchange medium flowing out from the heat exchange pieces **200**. A material of the current collecting piece **100** is not limited to plastic, aluminum or other metal pieces.

[0101] The second flow channel **101** refers to a channel that is disposed in the current collecting piece **100** and available for flow of the heat exchange medium.

[0102] The second joint portion **102** refers to a portion that is disposed on the current collecting piece **100** for being connected to the heat exchange piece **200**.

[0103] The plurality of second joint portions **102** are configured to be connected to the first joint portions **202** of the plurality of heat exchange pieces **200** respectively, such that the second flow channel **101** of the current collecting piece **100** communicates with the first flow channels **201** of the plurality of heat exchange pieces **200**. Understandably, the second joint portions **102** are connected to the first joint portions **202** in one-to-one correspondence. On the one hand, the current collecting piece **100** can be connected and fixed to the heat exchange pieces **200**; and on the other hand, after the first joint portions **202** and the second joint portions **102** are connected, the first flow channels **201** can further communicate with the second flow channel **101**, such that the heat exchange medium input into the second flow channel **101** flows into the first flow channels **201**, or the heat exchange medium in the first flow channels **201** flows into the second flow channel **101** to be collected and then output.

[0104] According to the heat exchange assembly **1** in this embodiment of this application, when the heat exchange assembly **1** is assembled with the battery units **500**, the plurality of battery units **500** are combined with the plurality of heat exchange pieces **200** first, and then the plurality of second joint portions **102** on the current collecting piece **100** are connected to the first joint portions **202** on the heat exchange pieces **200** in one-to-one correspondence, thereby completing assembly of the battery units **500** and the heat exchange assembly **1**. The heat exchange assembly **1** may simultaneously connect the plurality of heat exchange pieces **200** to the current collecting piece **100** through the second joint portions by arranging the plurality of second joint portions **102** on one current collecting piece **100**, which, compared with a form of separate connection of a plurality of current collecting pieces **100**, can save parts and reduce assembly procedures, thereby improving the assembly and fabrication efficiency.

[0105] In this embodiment, in the assembly process, the plurality of battery units **500** and the plurality of heat exchange pieces **200** are combined into an entirety first, and then the plurality of second joint portions **102** on the current collecting piece **100** are connected to the first joint portions **202** on the heat exchange pieces **200** in one-to-one correspondence. The integral connection manner has simpler and more convenient assembly operation and higher assembly efficiency than the assembly manner of separate connection of the plurality of current collecting pieces **100**.

[0106] A specific assembly method may be that every three heat exchange pieces **200** and two groups of battery units **500** form a unit module, a plurality of unit modules are arrayed and combined and then fabricated with the current collecting piece **100** and put into a box together to be fabricated into a battery. The battery units **500** are not limited to be prismatic or pouch-type. The direction of electrode terminals of the battery units **500** may be led out according to a grouping manner.

[0107] In another embodiment, with reference to FIG. **6**, FIG. **7** and FIG. **10**, the current collecting piece **100** includes a first current collecting piece and a second current collecting piece. The first current collecting piece and the second current collecting piece are located at two ends of the heat exchange pieces **200** respectively. The first current collecting piece is provided with a liquid inlet joint **610**, and the second current collecting piece is provided with a liquid outlet joint **620**. An external heat exchange medium enters a second flow channel **101** of the first current collecting piece from the liquid inlet joint **610**, and then flows into the first flow channels **201** through the second flow channel **101**. The heat exchange medium exchanges heat with the battery units **500** when flowing in the first flow channels **201**, and then the heat exchange medium in the first flow channels **201** flows into a second flow channel **101** of the second current collecting piece to be collected, and finally flows out from the liquid outlet joint **620**, thereby realizing heat exchange of the battery units **500**. The heat exchange medium in the first current collecting piece is simultaneously distributed to the heat exchange pieces **200** through multiple channels. Multi-channel distribution has the advantage of increasing the heat exchange rate between the heat exchange medium and the battery units **500**, such that the heating and cooling efficiency of the battery units **500** is further improved.

[0108] It is hereby noted that when the temperature of the heat exchange medium flowing in the first flow channels **201** is higher than that of the battery units **500**, heat of the heat exchange medium will be transferred to the battery units **500** through the heat exchange pieces **200**, serving to heat the battery units **500**. When the temperature of the heat exchange medium flowing in the second flow channel **101** is lower than that of the battery units **500**, heat of the battery units **500** will be transferred to the heat exchange medium through the heat exchange pieces **200**. After the heat exchange medium flows out from the heat exchange pieces **200**, the heat of the battery units **500** is taken away, thereby cooling the battery units **500**. The heat exchange medium may be water, air, cooling liquid, or the like.

[0109] In another embodiment of this application, with reference to FIG. **7**, FIG. **10** and FIG. **11**,

the current collecting piece **100** of the provided heat exchange assembly **1** includes a connecting face **103** oriented toward the heat exchange pieces **200**, and the plurality of second joint portions **102** are disposed on the connecting face **103**.

[0110] The connecting face **103** refers to a surface of the current collecting piece **100** oriented toward the heat exchange pieces **200**.

[0111] According to the heat exchange assembly **1** in this embodiment of this application, the plurality of second joint portions **102** are disposed on the connecting face **103**. In this way, during assembly, the connecting face **103** of the current collecting piece **100** is placed toward the heat exchange pieces **200**, and the second joint portions **102** are disposed opposite to the heat exchange pieces **200**, thereby facilitating the connection of the first joint portions **202** and the second joint portions **102**, and also contributing to the improvement of the assembly efficiency. In addition, the plurality of second joint portions **102** are located on the same connecting face **103**, and the structure of the current collecting piece **100** is also relatively simple, which facilitates machining and manufacturing.

[0112] In another embodiment of this application, with reference to FIG. 7, FIG. **11** and FIG. **14**, the plurality of heat exchange pieces **200** of the provided heat exchange assembly **1** are spaced apart along a first direction (referring to a direction indicated by an arrow X), and the plurality of second joint portions **102** are spaced out on the connecting face **103** along the first direction, such that the plurality of second joint portions **102** are in one-to-one correspondence with the first joint portions **202** of the plurality of heat exchange pieces **200**.

[0113] The first direction refers to an arrangement direction in which the plurality of heat exchange pieces **200** and the plurality of groups of battery units **500** are alternately stacked.

[0114] The plurality of heat exchange pieces **200** are spaced apart along the first direction.

Understandably, the first joint portions **202** on the plurality of heat exchange pieces **200** are also spaced apart along the first direction. In addition, an accommodation clearance **203** is formed between two adjacent heat exchange pieces **200**, and the accommodation clearance **203** provides a mounting space for a group of battery units **500**. The adjacent heat exchange pieces **200** exchange heat with two opposite sides of the group of battery units **500**, such that double-sided heat exchange of the battery units **500** is realized.

[0115] According to the heat exchange assembly **1** in this embodiment of this application, both the plurality of heat exchange pieces **200** and the plurality of second joint portions **102** on the current collecting piece **100** are distributed at intervals along the first direction, and the plurality of second joint portions **102** are in one-to-one correspondence with the first joint portions **202** on the plurality of heat exchange pieces **200**. In this way, when the heat exchange assembly **1** is assembled with the battery units **500**, the plurality of second joint portions **102** on the current collecting piece **100** are adjusted to be distributed along the first direction and opposite to the first joint portions **202**, such that the first joint portions **202** can be aligned with the second joint portions **102**, which makes the connecting operation of the heat exchange piece and the current collecting piece **100** simpler and quicker, and also contributes to the improvement of the assembly efficiency.

[0116] In another embodiment of this application, with reference to FIG. 7, FIG. **10** and FIG. **11**, the provided heat exchange assembly **1** further includes connectors **300**. The connector **300** is disposed at junction of the first joint portion **202** and the second joint portion **102** to connect the first joint portion **202** to the second joint portion **102**.

[0117] The connector **300** refer to a component for connecting the first joint portion **202** to the second joint portion **102**.

[0118] According to the heat exchange assembly **1** in this embodiment of this application, the first joint portion **202** and the second joint portion **102** are connected through the connector **300**, such that the connecting operation of the first joint portion **202** and the second joint portion **102** is simple and quick, which contributes to the improvement of the assembly efficiency of the heat exchange assembly **1** and the battery units **500**.

[0119] In another embodiment of this application, with reference to FIG. 10, FIG. 12 and FIG. 14, the first joint portion **202** of the provided heat exchange assembly **1** includes a first through hole **2021**. The second joint portion **102** includes a second through hole **1021**. The connector **300** includes a first end and a second end that extend along a connecting direction (referring to a direction indicated by an arrow Y) of the first joint portion **202** and the second joint portion **102**. The first end is connected into the first through hole **2021**. The second end is connected into the second through hole **1021**. The connector **300** includes a connecting hole **301** running through the first end and the second end.

[0120] The first through hole **2021** refers to a hole structure that is formed at and runs through the first joint portion **202**, and communicates with the first flow channel **201**.

[0121] The second through hole **1021** refers to a hole structure that is formed at and runs through the second joint portion **102**, and communicates with the first flow channel **201**.

[0122] The first end refers to an end of the connector **300** that is connected to the first through hole **2021**.

[0123] The second end refers to an end of the connector **300** that is connected to the second through hole **1021**.

[0124] The connecting hole **301** refers to a hole structure that is formed in the connector **300** and runs through the first end and the second end.

[0125] According to the heat exchange assembly **1** in this embodiment of this application, the first end and the second end of the connector **300** are connected into the first through hole **2021** and the second through hole **1021** respectively. Connecting structures between the connector **300** and the first joint portion **202** as well as between the connector **300** and the second joint portion **102** are simple, and the connecting operation is also simpler, which contributes to the improvement of the assembly efficiency of the heat exchange assembly **1** and the battery units **500**. In addition, the connector **300** includes the connecting hole **301** running through the first end and the second end. After the first end and the second end of the connector **300** are connected into the first through hole **2021** and the second through hole **1021** respectively, the connecting hole **301** enables the first flow channel **201** to communicate with the second flow channel **101**, such that the current collecting piece **100** communicates with the heat exchange piece. In this way, after the first end and the second end of the connector **300** are connected into the first through hole **2021** and the second through hole **1021** respectively, the connection, fixing and communication of the current collecting piece **100** and the heat exchange piece may be simultaneously realized, thereby simplifying assembly operation, which contributes to the improvement of the assembly efficiency of the heat exchange assembly **1** and the battery units **500**.

[0126] In another embodiment of this application, with reference to FIG. 10, FIG. 11 and FIG. 12, the first ends of the provided heat exchange assembly **1** are inserted in the first through holes **2021**, and the second ends are inserted in the second through holes **1021**.

[0127] According to the heat exchange assembly **1** in this embodiment of this application, the first end is inserted in the first through hole **2021**, and the second end is inserted in the second through hole **1021**, such that the connector **300** is connected to the current collecting piece **100** and the heat exchange piece **200** respectively. The connecting operation is simple, which contributes to the improvement of the assembly efficiency of the heat exchange assembly **1** and the battery units **500**. In addition, the first end of the connector **300** is inserted in the first through hole **2021**, and the second end of the connector **300** is inserted in the second through hole **1021**, such that a distance between the current collecting piece **100** and the heat exchange piece **200** can be reduced, thereby reducing a space occupied by the heat exchange assembly **1** in the box and improving the energy intensity of the battery.

[0128] In this embodiment, before assembly, the connectors **300** may be pre-mounted on the heat exchange pieces **200**. During assembly, the second joint portions **102** on the current collecting piece **100** are directly in one-to-one correspondence with the connectors **300** respectively. In this

way, the number of parts during assembly can be reduced, fabrication procedures are reduced, and the assembly efficiency is further improved.

[0129] In another embodiment, the connector **300** may be a connecting pipe. Two ends of the connecting pipes are inserted in the first through holes **2021** and the second through holes **1021** respectively, and can communicate with the first flow channels **201** and the second flow channel **101**, such that the mounting operation of the heat exchange pieces **200** and the current collecting piece **100** is simple, which contributes to the improvement of the assembly efficiency. The connecting pipes may be plastic pipes, or may be metal pipes made by metals, such as aluminum.

[0130] In another embodiment of this application, with reference to FIG. **10**, the provided heat exchange assembly **1** further includes first seals **410**. The first seal **410** is disposed between the connector **300** and a hole wall of the first through hole **2021**; and/or, the first seal **410** is disposed between the connector **300** and a hole wall of the second through holes **1021**.

[0131] The first seal **410** refers to a component capable of realizing a sealed connection. The first seal **410** may be a sealing ring, a sealing washer, or the like.

[0132] Understandably, the first seal **410** is disposed between the connector **300** and the hole wall of the first through hole **2021**. The first seal **410** can realize a sealed connection between the connector **300** and the first through hole **2021**, thereby reducing the risk of leakage of the heat exchange medium from between the connectors **300** and the holes wall of the first through hole **2021**, which contributes to the improvement of the heat exchange efficiency.

[0133] Or, with reference to FIG. **7**, the first seal **410** is disposed between the connector **300** and the hole wall of the second through hole **1021**. The first seal **410** realize a sealed connection between the connector **300** and the second through hole **1021**, thereby reducing the risk of leakage of the heat exchange medium from between the connector **300** and the hole wall of the second through hole **1021**, which contributes to the improvement of the heat exchange efficiency.

[0134] Or, the first seals **410** are disposed between the connector **300** and the hole wall of the first through hole **2021** as well as between the connector **300** and the hole wall of the second through hole **1021** respectively. The sealed connections between the connector **300** and the hole wall of the first through hole **2021** as well as between the connector **300** and the hole wall of the second through hole **1021** are realized through the first seals **410** respectively, thereby reducing the risk of leakage of the heat exchange medium from between the connector **300** and the hole wall of the first through hole **2021** as well as between the connector **300** and the hole wall of the second through hole **1021**, which more contributes to the improvement of the heat exchange efficiency.

[0135] In another embodiment of this application, with reference to FIG. **11**, FIG. **12** and FIG. **13**, the heat exchange piece **200** of the provided heat exchange assembly **1** includes a first side plate **210** and a second side plate **220**. The first side plate **210** and the second side plate **220** are connected and close in to form the first flow channel **201** and the first through hole **2021**.

[0136] The first side plate **210** refers to a plate on the heat exchange piece **200** that exchanges heat with one of two adjacent groups of battery units **500**.

[0137] The second side plate **220** refers to a plate on the heat exchange piece **200** that exchanges heat with the other of the two adjacent groups of battery units **500**.

[0138] The heat exchange piece **200** according to this embodiment of this application is formed by the connecting and closing in of the first side plate **210** and the second side plate **220**, and is simple in structure and also convenient to machine and manufacture. In addition, in a specific manufacturing process, the first side plate **210** and the second side plate **220** may be manufactured by stamping, such that the machining and manufacturing of the heat exchange piece **200** are simpler, and the manufacturing cost is lower.

[0139] In another embodiment, with reference to FIG. **11** and FIG. **12**, connecting bulges **211** are disposed at circumferences of surfaces opposite to each other of the first side plate **210** and the second side plate **220** respectively, and the two connecting bulges **211** are bonded or welded together, such that the first side plate **210** and the second side plate **220** can close in to form the

first flow channel **201** therebetween. A first semi-ring bulge **212** is disposed on a side portion of the first side plate **210** oriented toward the current collecting piece **100**. A second semi-ring bulge **221** is disposed on a side portion of the second side plate **220** oriented toward the current collecting piece **100**. The first semi-ring bulge **212** and the second semi-ring bulge **221** close in to form the first through hole **2021**.

[0140] In another embodiment of this application, with reference to FIG. **11**, FIG. **12** and FIG. **13**, the connector **300** of the provided heat exchange assembly **1** includes a first connecting portion **310** and a second connecting portion **320**. The first connecting portion **310** is connected to the first side plate **210**. The second connecting portion **320** is connected to the second side plate **220**. The first connecting portion **310** and the second connecting portion **320** close in to form the connecting hole.

[0141] The first connecting portion **310** refers to a portion on the connector **300** that is connected to the first side plate **210**.

[0142] The second connecting portion **320** refers to a portion on the connector **300** that is connected to the second side plate **220**.

[0143] According to the heat exchange assembly **1** in this embodiment of this application, the connector **300** uses a separate structure of the first connecting portion **310** and the second connecting portion **320**, and the first connecting portion **310** and the second connecting portion **320** may be manufactured by stamping, or the like, such that the machining and manufacturing of the connector **300** are simpler. Certainly, in other embodiments, the connector **300** may also be of an integral tubular structure, or the like.

[0144] In another embodiment of this application, with reference to FIG. **11**, FIG. **12** and FIG. **13**, the first side plate **210** and the first connecting portion **310** of the provided heat exchange assembly **1** are of a one-piece structure. Or, the first connecting portion **310** is bonded to the first side plate **210**. Or, the first connecting portion **310** is welded to the first side plate **210**.

[0145] The first side plate **210** and the first connecting portion **310** are of the one-piece structure. Understandably, the first side plate **210** and the first connecting portion **310** may be of the one-piece structure manufactured by a one-piece process of extrusion, injection molding, die-casting, or the like, which can reduce machining process steps and reduce the manufacturing cost. In addition, the first side plate **210** and the first connecting portion **310** are of the one-piece structure, which contributes to the improvement of the structural strength of the heat exchange piece **200**.

[0146] The first connecting portion **310** is bonded to the first side plate **210**. Understandably, the first connecting portion **310** is bonded to the first side plate **210** through a binder. The connecting operation of the first side plate **210** and the first connecting portion **310** is simple and convenient, which contributes to reducing the manufacturing cost of the heat exchange piece. The binder may be a structural adhesive, super glue, or the like.

[0147] The first connecting portion **310** is welded to the first side plate **210**. Understandably, the first connecting portion **310** is fixed to the first side plate **210** by welding. The connecting operation of the first side plate **210** and the first connecting portion **310** is simple and convenient, which contributes to reducing the manufacturing cost of the heat exchange piece.

[0148] In another embodiment of this application, with reference to FIG. **11**, FIG. **12** and FIG. **13**, the second side plate **220** and the second connecting portion **320** of the provided heat exchange assembly **1** are of a one-piece structure. Or, the second connecting portion **320** is bonded to the second side plate **220**. Or, the second connecting portion **320** is welded to the second side plate **220**.

[0149] The second side plate **220** and the second connecting portion **320** are of the one-piece structure. Understandably, the second side plate **220** and the second connecting portion **320** may be of the one-piece structure manufactured by a one-piece process of extrusion, injection molding, die-casting, or the like, which can reduce machining process steps and reduce the manufacturing cost. In addition, the second side plate **220** and the second connecting portion **320** are of the one-piece structure, which contributes to the improvement of the structural strength of the heat

exchange piece **200**.

[0150] The second connecting portion **320** is bonded to the second side plate **220**. Understandably, the second connecting portion **320** is bonded to the second side plate **220** through a binder. The connecting operation of the second side plate **220** and the second connecting portion **320** is simple and convenient, which contributes to reducing the manufacturing cost of the heat exchange piece. The binder may be a structural adhesive, super glue, or the like.

[0151] The second connecting portion **320** is welded to the second side plate **220**. Understandably, the second connecting portion **320** is fixed to the second side plate **220** by welding. The connecting operation of the second side plate **220** and the second connecting portion **320** is simple and convenient, which contributes to reducing the manufacturing cost of the heat exchange piece.

[0152] In another embodiment of this application, with reference to FIG. **15**, FIG. **16** and FIG. **17**, the second end of the provided heat exchange assembly **1** is threaded in the second through hole **1021**, and the first end is screwed in the first through hole **2021**, such that the heat exchange piece **200** is connected to the current collecting piece **100**.

[0153] The first end is screwed in the first through hole **2021**. Understandably, the first end is externally provided with external threads. The hole wall of the first through hole **2021** is provided with internal threads. Through threaded fit of the external threads and the internal threads, the first end is fixed into the first through hole **2021**.

[0154] According to the heat exchange assembly **1** in this embodiment of this application, the second end of the connector **300** is threaded in the second through hole **1021**, and the first end of the connector **300** is screwed in the first through hole **2021**. By threaded connection, on the one hand, the connecting operation is simple, which contributes to the improvement of the assembly effect; and in addition, the manner of threaded connection has good connection reliability, ensures stable connection between the connector **300** and the heat exchange piece **200** and also contributes to the improving the mechanical strength of the heat exchange assembly **1**, thereby improving the structural strength of the battery **1100**. Moreover, threaded connection also has a sealing effect, which can also reduce the leakage of the heat exchange medium and improve the heat exchange effect.

[0155] In a specific embodiment, the connectors **300** may be fasteners. The connecting hole **301** is formed in the fastener. The fasteners may be screws, bolts, cap screws, or the like. The heat exchange piece **200** and the current collecting piece **100** are connected by the fasteners such as the screws, the bolts, or the cap screws. The connecting operation between the heat exchange piece **200** and the current collecting piece **100** is simpler, and the connection reliability is good.

[0156] In another embodiment of this application, with reference to FIG. **17**, FIG. **18** and FIG. **19**, the current collecting piece **100** of the provided heat exchange assembly **1** includes a first plate body **110** and a second plate body **120**. The first plate body **110** is connected to the second plate body **120**. The first plate body **110** and the second plate body **120** close in to form the second flow channel **101**. The second through holes **1021** are disposed on the first plate body **110**.

[0157] The first plate body **110** refers to a plate that is disposed oriented toward the heat exchange pieces **200** and provided with the second through holes **1021**.

[0158] The second plate body **120** refers to a plate that is disposed oriented back from the heat exchange pieces **200**.

[0159] The first plate body **110** is connected to the second plate body **120**. Understandably, the first plate body **110** and the second plate body **120** may be connected through a screw structure, a buckle structure, an adhesive structure, a magnetic adsorption structure, or other structures capable of connecting and fixing the two components.

[0160] The first plate body **110** and the second plate body **120** close in to form the second flow channel **101**. Understandably, a surface of the first plate body **110** oriented toward the second plate body **120** is recessed toward an inside of the first plate body **110** to form a recessed space, and an inner wall of the recessed space and a surface of the second plate body **120** oriented toward the first

plate body **110** jointly close in to form the second flow channel **101**. Or, a surface of the second plate body **120** oriented toward the first plate body **110** is recessed toward an inside of the second plate body **120** to form a recessed space, and an inner wall of the recessed space and the surface of the first plate body **110** oriented toward the second plate body **120** jointly close in to form the second flow channel **101**. Or, the surface of the first plate body **110** oriented toward the second plate body **120** is recessed inward to form the first recessed space, the surface of the second plate body **120** oriented toward the first plate body **110** is recessed toward the inside of the second plate body **120** to form the second recessed space, and the inner wall of the first recessed space and the inner wall of the second recessed space jointly close in to form the second flow channel **101**.

[0161] According to the heat exchange assembly **1** in this embodiment of this application, when the first joint portion **202** is connected to the second joint portion **102**, the first end of the connector **300** passes through the second through hole **1021** first, and then is threadedly locked in the first through hole **2021** to fix the first plate body **110** on the heat exchange piece **200**, and finally the second plate body **120** is fixed on the first plate body **110**, thereby realizing assembly of the heat exchange piece **200** and the current collecting piece **100**. The assembly operation is simple, convenient and quick, which contributes to the improvement of the heat exchange assembly **1** and the battery units **500**.

[0162] In another embodiment of this application, with reference to FIG. **17**, FIG. **20** and FIG. **21**, a second seal **420** is sandwiched between the first plate body **110** and the second plate body **120** of the provided heat exchange assembly **1**, and the second seal **420** surrounds the second flow channel **101**. And/or, one of the first plate body **110** and the second plate body **120** is provided with an annular seal bulge **111**, and the other of the first plate body **110** and the second plate body **120** is provided with an annular seal groove **121**. The annular seal bulge **111** is inserted in the annular seal groove **121** and surrounds the second flow channel **101**.

[0163] The second seal **420** refers to a sealing component that is sandwiched between the first plate body **110** and the second plate body **120**. The second seal **420** may be a sealing ring, a sealing washer, or the like.

[0164] In an embodiment, the second seal **420** is sandwiched between the first plate body **110** and the second plate body **120**, and the second seal **420** surrounds the second flow channel **101**. Understandably, the second seal **420** is sandwiched between the first plate body **110** and the second plate body **120** which are provided around the second flow channel **101**. In this way, when flowing outward from between the first plate body **110** and the second plate body **120**, the heat exchange medium in the second flow channel **101** will be blocked by the second seal **420**, thereby reducing the leakage of the heat exchange medium, and improving sealing performance of the current collecting piece **100**, which contributes to the improvement of the heat exchange effect.

[0165] In an embodiment, one of the first plate body **110** and the second plate body **120** is provided with the annular seal bulge **111**, and the other of the first plate body **110** and the second plate body **120** is provided with the annular seal groove **121**. The annular seal bulge **111** is inserted in the annular seal groove **121**, and the annular seal bulge **111** surrounds the second flow channel **101**. Understandably, the annular seal bulge **111** is disposed on the first plate body **110**, and the annular seal groove **121** is disposed on the second plate body **120**. Or, the annular seal bulge **111** is disposed on the second plate body **120**, and the annular seal groove **121** is disposed on the first plate body **110**. For the first plate body **110** and the second plate body **120**, after the annular seal bulge **111** is inserted in the annular seal groove **121**, the heat exchange medium in the second flow channel **101** flows to the annular seal bulge **111** from a clearance between the first plate body **110** and the second plate body **120** and then is blocked by the annular seal bulge **111**, thereby reducing the risk of leakage of the heat exchange medium from between the first plate body **110** and the second plate body **120**, and improving the sealing performance of the current collecting piece **100**, which contributes to the heat exchange effect.

[0166] In an embodiment, the annular seal groove **121** may be filled with a sealant. A clearance

between the annular seal groove **121** and the annular seal bulge **111** may be completely filled with the sealant. A better sealing effect can be realized, the risk of leakage of the heat exchange medium in the second flow channel **101** is lower, and the sealing performance of the current collecting piece **100** is better.

[0167] In an embodiment, the second seal **420** is sandwiched between the first plate body **110** and the second plate body **120** of the heat exchange assembly **1**, and the second seal **420** surrounds the second flow channel **101**. One of the first plate body **110** and the second plate body **120** is provided with the annular seal bulge **111**, and the other of the first plate body **110** and the second plate body **120** is provided with the annular seal groove **121**. The annular seal bulge **111** is inserted in the annular seal groove **121**, and the annular seal bulge **111** surrounds the second flow channel **101**. Understandably, the second seal **420** as well as the annular seal bulge **111** and the annular seal groove **121** are used together between the first plate body **110** and the second plate body **120** for sealing of the second flow channel **101**. In this way, the sealing performance of the second flow channel **101** is better, the risk of leakage of the heat exchange medium is lower, and the sealing performance of the current collecting piece **100** is better. the second seal **420** may be located between the annular seal bulge **111** and the second flow channel **101**, or the annular seal bulge **111** is located between the second flow channel **101** and the second seal **420**, which is specifically selected according to actual needs, and is not limited herein.

[0168] In another embodiment of this application, with reference to FIG. **10**, FIG. **17**, FIG. **20** and FIG. **21**, the second joint portion **102** of the provided heat exchange assembly **1** further includes a first bulge **1022**. The first bulge **1022** is disposed protruding from the connecting face **103**. The second through hole **1021** runs through the first bulge **1022** along the connecting direction of the first joint portion **202** and the second joint portion **102**.

[0169] The first bulge **1022** refers to a portion on the current collecting piece **100** that protrudes from the connecting face **103**.

[0170] According to the heat exchange assembly **1** in this embodiment of this application, the disposing of the first bulge **1022** can increase the thickness of the second through hole **1021**, improve the structural strength of the second through hole **1021**, and improve the connection reliability of the current collecting piece **100** and the connector **300**. In addition, the disposing of second bulge **2022** can also increase the depth of the second through hole **1021**, thereby increasing the connection area **103** of the connector **300** and the second through hole **1021**, improving the connection reliability of the current collecting piece **100** and the connector **300**, and in turn, improving the structural strength of the entire heat exchange assembly **1**.

[0171] In another embodiment of this application, with reference to FIG. **10** and FIG. **17**, the first joint portion **202** of the provided heat exchange assembly **1** further includes a second bulge **2022**. The second bulge **2022** is disposed on an outer surface of the heat exchange piece **200** oriented toward the current collecting piece **100**. The first through hole **2021** runs through the second bulge **2022** along the connecting direction of the first joint portion **202** and the second joint portion **102**. The first bulge **1022** abuts against the second bulge **2022**.

[0172] The second bulge **2022** refers to a portion on the heat exchange piece that protrudes from the outer surface of the heat exchange piece **200** oriented toward the current collecting piece **100**.

[0173] According to the heat exchange assembly **1** in this embodiment of this application, the disposing of the second bulge **2022** can increase the thickness of the first through hole **2021**, improve the structural strength of the first through hole **2021**, improve the connection reliability of the heat exchange piece **200** and the connector **300**, and in turn, improve the structural strength of the entire heat exchange assembly **1**. In addition, the first bulge **1022** abuts against the second bulge **2022**, which can limit a distance between the current collecting piece **100** and the heat exchange piece **200**. Then, the current collecting piece **100** has the same or nearly the same distance from each heat exchange piece **200**. The current collecting piece **100** has a flat and straight overall shape. Therefore, accurate mounting between the current collecting piece **100** and each heat

exchange piece **200** can be realized.

[0174] In another embodiment of this application, with reference to FIG. **17**, a third seal **430** is sandwiched between the first bulge **1022** and the second bulge **2022** of the provided heat exchange assembly **1**. The connector **300** is threaded in the third seal **430**.

[0175] The third seal **430** refers to a sealing component that is sandwiched between the first bulge **1022** and the second bulge **2022**. The third seal **430** may be a sealing ring, a sealing washer, or the like.

[0176] According to the heat exchange assembly **1** in this embodiment of this application, the third seal **430** is sandwiched between the first bulge **1022** and the second bulge **2022**. The connector **300** is inserted in the third seal **430**. In this way, when flowing to between the first bulge **1022** and the second bulge **2022**, the heat exchange medium flowing from between the connector **300** and the hole wall of the first through hole **2021** will be blocked by the third seal **430** and cannot continue flowing outward, which reduces the risk of leakage of the heat exchange medium in the first flow channel **201**. Similarly, the heat exchange medium flowing from between the connector **300** and the hole wall of the second through hole **1021** will also be blocked by the third seal **430** and cannot continue flowing outward, which reduces the risk of leakage of the heat exchange medium in the second flow channel **101**. The leakage of the heat exchange medium in the heat exchange assembly **1** is reduced, which contributes to the improvement of the heat exchange efficiency.

[0177] In another embodiment of this application, with reference to FIG. **10**, the second joint portion **102** of the provided heat exchange assembly **1** further includes a third bulge **1023**. The third bulge **1023** is disposed on an inner wall of the second flow channel **101** close to the heat exchange piece **200**. The second through hole **1021** runs through the third bulge **1023** along the connecting direction of the first joint portion **202** and the second joint portion **102**.

[0178] The third bulge **1023** refer to a portion on the current collecting piece **100** that protrudes from the inner wall of the second flow channel **101** close to the heat exchange piece **200**.

[0179] According to the heat exchange assembly **1** in this embodiment of this application, the disposing of the third bulge **1023** can increase the thickness of the second through hole **1021**, improve the structural strength of the second through hole **1021**, and improve the connection reliability of the current collecting piece **100** and the connector **300**. In addition, the disposing of third bulge **1023** can also increase the depth of the second through hole **1021**, thereby increasing the connection area **103** of the connector **300** and the second through hole **1021**, improving the connection reliability of the current collecting piece **100** and the connector **300**, and in turn, improving the structural strength of the entire heat exchange assembly **1**.

[0180] In another embodiment, with reference to FIG. **8**, FIG. **9** and FIG. **10**, the second bulge **2022** is disposed on the heat exchange piece **200**. The first bulge **1022** and the third bulge **1023** are disposed on the current collecting piece **100**. The first end of the connector **300** is inserted in the first through hole **2021**. The second end of the connector **300** is inserted in the second through hole **1021**. The second bulge **2022** can improve the structural strength of the first through hole **2021**, and improve the connection reliability of the heat exchange piece **200** and the first end. In addition, the first bulge **1022** and the third bulge **1023** can improve the structural strength of the second through hole **1021**, can increase the depth of the second through hole **1021**, and can increase the depth of the second end inserted in the second through hole **1021**, thereby improving the connection strength of the second end and the current collecting piece **100** as well as the connection reliability between the current collecting piece **100** and the heat exchange piece **200**. A seal accommodation groove **204** extending along a circumferential direction is formed in the hole wall of the first through hole **2021**. The first seal **410** is received in the seal accommodation groove **204**. The connector **300** is inserted in the first through hole **2021** and extrudes the second seal **420** to deform it, such that the second seal **420** seals a clearance between the connector **300** and the hole wall of the second through hole **1021**, thereby realizing a sealed connection of the connector **300** and the current collecting piece **100**.

[0181] With reference to FIG. 11 and FIG. 12, the heat exchange piece **200** includes a first side plate **210** and a second side plate **220**. Connecting bulges **211** are disposed at circumferences of surfaces opposite to each other of the first side plate **210** and the second side plate **220** respectively, and the two connecting bulges **211** are bonded or welded together, such that the first side plate **210** and the second side plate **220** can close in to form the first flow channel **201** therebetween. A first semi-ring bulge **212** is disposed on a side portion of the first side plate **210** oriented toward the current collecting piece **100**. A second semi-ring bulge **221** is disposed on a side portion of the second side plate **220** oriented toward the current collecting piece **100**. The first semi-ring bulge **212** and the second semi-ring bulge **221** close in to form the first through hole **2021**. The first connecting portion **310** is bonded or welded to the first semi-ring bulge **212**. The second connecting portion **320** is bonded or welded to the second semi-ring bulge **221**. After the first side plate **210** and the second side plate **220** are connected together, the first connecting portion **310** and the second connecting portion **320** are also connected and close in to form the connecting hole **301**. The first semi-ring bulge **212** and the second semi-ring bulge **221** jointly form the second bulge **2022**.

[0182] In another embodiment, with reference to FIG. 13, the first connecting portion **310** and the first side plate **210** are integrally molded by stamping. The second connecting portion **320** and the second side plate **220** are integrally molded by stamping. In this way, the heat exchange pieces **200** have a simple manufacturing manner, which contributes to the reduction of the manufacturing cost.

[0183] In another embodiment, with reference to FIG. 17, FIG. 20 and FIG. 21, the connectors **300** are first cap screws. The connecting hole **301** is formed in the first cap screw along an axial direction. The second through holes **1021** are formed in the first plate body **110**. The first through holes **2021** are formed in the heat exchange piece **200**. The first cap screw pass through the second through hole **1021** and then screwed in the first through hole **2021**. Then, the second plate body **120** is fixed to the first plate body **110** through second screws **440**. The heat exchange pieces **200** and the current collecting piece **100** are fixed by cap screws. The connecting operation is simple and quick, and stable and reliable connection of the heat exchange pieces **200** and the current collecting piece **100** can also be realized. The second screws **440** are plural in number and spaced apart along a circumference of the first plate body **110**, such that stable and reliable connection of the first plate body **110** and the second plate body **120** can be realized.

[0184] The second bulge **2022** is disposed on the heat exchange piece **200**. The first bulges **1022** are disposed on the first plate body **110**. Stepped groove is formed in surface of the first bulge **1022** oriented toward the second bulge **2022**. The second seal **420** is located in the stepped groove. In addition, the second bulge **2022** is inserted in the stepped groove and presses against the second seal **420**, such that the second seal **420** deforms, and clearance between the first bulge **1022** and the second bulge **2022** is sealed through the deformation of the second seal **420**, thereby reducing the leakage of the heat exchange medium and improving the sealing performance.

[0185] The first plate body **110** is provided with an annular abutting bulge **112**. The second plate body **120** is provided with an annular accommodation groove **122**. The second seal **420** is located in the annular accommodation groove **122**. The annular abutting bulge **112** is inserted in the annular accommodation groove **122** and presses against the second seal **420**, such that the second seal **420** deforms to seal a clearance between the annular abutting bulge **112** and the annular accommodation groove **122**, thereby sealing the second flow channel **101**.

[0186] The first plate body **110** is provided with an annular seal bulge **111**. The second plate body **120** is provided with an annular seal groove **121**. The annular seal bulge **111** surrounds the annular abutting bulge **112**. The annular seal bulge **111** is inserted in the annular seal groove **121**, thereby improving the sealing performance of the second flow channel **101** and reducing the leakage of the heat exchange medium.

[0187] In another embodiment of this application, a battery is provided, including the foregoing heat exchange assembly **1**.

[0188] The battery according to this embodiment of this application uses the foregoing heat exchange assembly **1**. The assembly operation of the heat exchange assembly **1** and battery units **500** is simple, and the assembly efficiency is high, which contributes to the improvement of the production efficiency of the battery and the reduction of the manufacturing cost of the battery.

[0189] The battery according to this embodiment of this application employs all the technical solutions of all the preceding embodiments, and therefore, achieves all the beneficial effects brought by the technical solutions of the preceding embodiments, the details of which are omitted here.

[0190] In another embodiment of this application, an electrical device is provided, including the foregoing battery.

[0191] The electrical device according to this embodiment of this application uses the foregoing battery. The manufacturing cost of the battery is low, which contributes to the reduction of the manufacturing cost of the electrical device.

[0192] The electrical device according to this embodiment of this application employs all the technical solutions of all the preceding embodiments, and therefore, achieves all the beneficial effects brought by the technical solutions of the preceding embodiments, the details of which are omitted here.

[0193] The above are only optional embodiments of this application and are not intended to limit this application. Any modifications, equivalent replacements, improvements, and the like made within the spirit and principles of this application shall fall within the protection scope of this application.

Claims

1. A heat exchange assembly, comprising: a plurality of heat exchange pieces, the heat exchange piece being constructed with a first flow channel and a first joint portion, and the first joint portion communicating with the first flow channel; and a current collecting piece, the current collecting piece being constructed with a second flow channel and a plurality of second joint portions, and the plurality of second joint portions each communicating with the second flow channel and being configured to be connected to the first joint portions of the plurality of heat exchange pieces respectively, such that the second flow channel of the current collecting piece communicates with the first flow channels of the plurality of heat exchange pieces.
2. The heat exchange assembly according to claim 1, wherein the current collecting piece comprises a connecting face oriented toward the heat exchange pieces, and the plurality of second joint portions are disposed on the connecting face.
3. The heat exchange assembly according to claim 2, wherein the plurality of heat exchange pieces are spaced apart along a first direction, and the plurality of second joint portions are spaced out on the connecting face along the first direction, such that the plurality of second joint portions are in one-to-one correspondence with the first joint portions of the plurality of heat exchange pieces.
4. The heat exchange assembly according to claim 2, wherein the heat exchange assembly further comprises connectors, and each connector is disposed at junction of the first joint portion and the second joint portion to connect the first joint portion to the second joint portion.
5. The heat exchange assembly according to claim 4, wherein the first joint portion comprises a first through hole, the second joint portion comprises a second through hole, the connector comprises a first end and a second end that extend along a connecting direction of the first joint portion and the second joint portion, the first end is connected into the first through hole, the second end is connected into the second through hole, and the connector comprises a connecting hole running through the first end and the second end.
6. The heat exchange assembly according to claim 5, wherein the first end is inserted in the first through hole, and the second end is inserted in the second through hole.

7. The heat exchange assembly according to claim 6, wherein the heat exchange assembly further comprises first seals, and the first seal is disposed between the connector and hole wall of the first through hole; and/or, the first seal is disposed between the connector and hole wall of the second through hole.
 8. The heat exchange assembly according to claim 6, wherein the heat exchange piece comprises a first side plate and a second side plate, and the first side plate and the second side plate are connected and close in to form the first flow channel and the first through hole.
 9. The heat exchange assembly according to claim 8, wherein the connector comprises a first connecting portion and a second connecting portion, the first connecting portion is connected to the first side plate, the second connecting portion is connected to the second side plate, and the first connecting portion and the second connecting portion close in to form the connecting hole.
 10. The heat exchange assembly according to claim 9, wherein the first side plate and the first connecting portion are of a one-piece structure; or the first connecting portion is bonded to the first side plate; or the first connecting portion is welded to the first side plate.
 11. The heat exchange assembly according to claim 9, wherein the second side plate and the second connecting portion are of a one-piece structure; or the second connecting portion is bonded to the second side plate; or the second connecting portion is welded to the second side plate.
 12. The heat exchange assembly according to claim 5, wherein the second end is threaded in the second through hole, and the first end is screwed in the first through hole, such that the heat exchange piece is connected to the current collecting piece.
 13. The heat exchange assembly according to claim 12, wherein the current collecting piece comprises a first plate body and a second plate body, and the first plate body and the second plate body are connected and close in to form the second flow channel; and the second through holes are disposed on the first plate body.
 14. The heat exchange assembly according to claim 13, wherein a second seal is sandwiched between the first plate body and the second plate body, and the second seal surrounds the second flow channel; and/or, one of the first plate body and the second plate body is provided with an annular seal bulge, and the other of the first plate body and the second plate body is provided with an annular seal groove; and the annular seal bulge is inserted in the annular seal groove and surrounds the second flow channel.
 15. The heat exchange assembly according to claim 5, wherein the second joint portion further comprises a first bulge, the first bulge protrudes from the connecting face, and the second through hole runs through the first bulge along the connecting direction of the first joint portion and the second joint portion.
 16. The heat exchange assembly according to claim 15, wherein the first joint portion further comprises a second bulge, the second bulge is disposed on an outer surface of the heat exchange piece oriented toward the current collecting piece, the first through hole runs through the second bulge along the connecting direction of the first joint portion and the second joint portion, and the first bulge abuts against the second bulge.
 17. The heat exchange assembly according to claim 16, wherein a third seal is sandwiched between the first bulge and the second bulge, and the connector is threaded in the third seal.
 18. The heat exchange assembly according to claim 5, wherein the second joint portion further comprises a third bulge, the third bulge is disposed on an inner wall of the second flow channel close to the heat exchange piece, and the second through hole runs through the third bulge along the connecting direction of the first joint portion and the second joint portion.
 19. A battery, comprising the heat exchange assembly according to claim 1.
 20. An electrical device, comprising the battery according to claim 19.
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